Eghbal A. Hosseini

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Eghbal A. Hosseini https://eghbalhosseini.github.io

Current Position

2024 – Present **Visiting Researcher**, Google DeepMind, San Francisco, CA

Education

2016 – 2024 **Ph.D. Computational Neuroscience**, Massachusetts Institute of Technology (MIT)
Thesis: *Towards Synergistic Understanding of Language Processing in Biological and Artificial Systems*.

2023 **Analytical Connectionism**, Gatsby Computational Neuroscience Unit,
University College London.

2017 **Brains, Minds, and Machines**, Marine Biological Laboratory
University of Chicago.

2012 – 2014 **M.Sc. Electrical Engineering**, George Mason University (GMU)
Thesis: *Multi-rate state-dependent primitives underlie the motor adaptation and unlearning to motion dependent force perturbation*.

2005 – 2010 **B.Sc. Electrical Engineering**, Iran University of Science and Technology (IUST)
Thesis: *Position control of DC motor using wavelet based multi-resolution analysis.*.

Research Publications

Selected

Hosseini, **E. A.**, Cheung, B., Fedorenko, E., & Williams, A. H. (2026). Modulating cross-modal convergence with single-stimulus, intra-modal dispersion. *ICLR 2026 Re-Align Workshop*. .

Hosseini, **E. A.**, Li, Y., Bahri, Y., Campbell, D., & Lampinen, A. K. (2026). Context structure reshapes the representational geometry of language models. *arXiv*. .

Hosseini, **E. A.**, Casto, C., Zaslavsky, N., Conwell, C., Richardson, M., & Fedorenko, E. (2024). Universality of representation in biological and artificial neural networks. *bioRxiv*. .

Hosseini, **E. A.**, Schrimpf, M., Zhang, Y., Bowman, S., Zaslavsky, N., & Fedorenko, E. (2024). Artificial neural network language models predict human brain responses to language even after a developmentally realistic amount of training. *Neurobiol Lang (Camb)*, *5*(1), 43–63. .

Hosseini, **E. A.**, & Fedorenko, E. (2023). Large language models implicitly learn to straighten neural sentence trajectories to construct a predictive representation of natural language. *NeurIPS*. .

Hosseini, **E. A.**, Zaslavsky, N., Casto, C., & Fedorenko, E. (2023). Teasing apart the representational spaces of ANN language models to discover key axes of model-to-brain alignment. *Computational Cognitive Neuroscience, (Oral presentation, top 5% submission)*. .

Aditional

Lampinen, A. K., Li, Y., **Hosseini**, **E. A.**, Bhardwaj, S., & Shanahan, M. (2026). Linear representations in language models can change dramatically over a conversation. *arXiv*.

Regev, T. I., Casto, C., **Hosseini**, **E. A.**, Adamek, M., Ritaccio, A. L., Willie, J. T., Brunner, P., & Fedorenko, E. (2024). Neural populations in the language network differ in the size of their temporal receptive windows. *Nat. Hum. Behav.*, *8*(10), 1924–1942. .

Schrimpf, M., Blank, I. A., Tuckute, G., Kauf, C., **Hosseini**, **E. A.**, Kanwisher, N., Tenenbaum, J. B., & Fedorenko, E. (2021). The neural architecture of language: Integrative modeling converges on predictive processing. *Proc. Natl. Acad. Sci. U. S. A.*, *118*(45). 🔗.

Tremblay, S., Acker, L., Afraz, A., Albaugh, D. L., Amita, H., Andrei, A. R., Angelucci, A., Aschner, A., Balan, P. F., Basso, M. A., Benvenuti, G., Bohlen, M. O., Caiola, M. J., Calcedo, R., Cavanaugh, J., Chen, Y., Chen, S., Chernov, M. M., Clark, A. M., … Platt, M. L. (2020). An open resource for non-human primate optogenetics. *Neuron*. 🔗.

Wang, J., **Hosseini**, **E. A.**, Meirhaeghe, N., Akkad, A., & Jazayeri, M. (2020). Reinforcement regulates timing variability in thalamus. *Elife*, *9*. 🔗.

Alhussein, L., **Hosseini**, **E. A.**, Nguyen, K. P., Smith, M. A., & Joiner, W. M. (2019). Dissociating effects of error size, training duration, and amount of adaptation on the ability to retain motor memories. *J. Neurophysiol.*, *122*(5), 2027–2042. 🔗.

Nguyen, K. P., Zhou, W., McKenna, E., Colucci-Chang, K., Bray, L. C. J., **Hosseini**, **E. A.**, Alhussein, L., Rezazad, M., & Joiner, W. M. (2019). The 24-h savings of adaptation to novel movement dynamics initially reflects the recall of previous performance. *J. Neurophysiol.*, *122*(3), 933–946. 🔗.

Remington, E. D., Narain, D., **Hosseini**, **E. A.**, & Jazayeri, M. (2018). Flexible sensorimotor computations through rapid reconfiguration of cortical dynamics. *Neuron*, *98*(5), 1005–1019.e5. 🔗.

Wang, J., Narain, D., **Hosseini**, **E. A.**, & Jazayeri, M. (2018). Flexible timing by temporal scaling of cortical responses. *Nat. Neurosci.*, *21*(1), 102–110. 🔗.

Hosseini, **E. A.**, Nguyen, K. P., & Joiner, W. M. (2017). The decay of motor adaptation to novel movement dynamics reveals an asymmetry in the stability of motion state-dependent learning. *PLoS Comput. Biol.*, *13*(5), e1005492. 🔗.

Hosseini, **E. A.**, & Sadjadian, H. (2015). Noise resistant design of wavelet based multiresolution control. *American Control Conference (ACC), 2015*, 4959–4963. 🔗.

Technical & Research Skills

Research Expertise	Computational neuroscience, Language modeling, Representation learning, Neural coding
Machine Learning Frameworks	PyTorch, TensorFlow, Slurm
Programming Languages	Python, MATLAB, R, Simulink, Pycharm, LaTeX
Web Development	HTML, CSS, JavaScript
Operating Systems	Linux, Macintosh OS, Microsoft Windows
Others	Neuroimaging analysis, Data visualization, Academic research, teaching, undergraduate advising, publishing and consultation
Languages	Fluent in reading, writing, and speaking *English* and *Persian*

Experiences & Awards

Research Experience

2025-Present	**Visiting Scientist**, Special Year on Large Language Models and Transformers, Part 2, Simons Institute for the Theory of Computing, UC Berkeley
2024-Present	**Postdoctoral Associate**, Dr. Evelina Fedorenko, EvLab, McGovern Institute for Brain Research, MIT
2019-2024	**Graduate Research Assistant**, Dr. Evelina Fedorenko, EvLab, McGovern Institute for Brain Research, MIT
2017-2018	**Graduate Fellowship Student**, Dr. Edward S. Boyden, Synthetic Neurbiology Group, McGovern Institute for Brain Research, MIT

Experiences & Awards (continued)

2015-2016 **Technical Assistant**, Dr. Mehrdad Jazayeri, JazLab, McGovern Institute for Brain Research, MIT

2013-2014 **Graduate Research Assistant**, Dr. Wilsaan Joiner, Sensorimotor Integration Lab, Volgenau School of Engineering, GMU

Teaching Experience

2020 **Teaching Assistant**, Dr. Michale Fee, Introduction to Neural Computation, Department of Brain and Cognitive Sciences, MIT

2017 **Teaching Assistant**, Dr. Tomaso Poggio, Science of Intelligence, MIT

Awards and Achievements

2020 **Friends of the McGovern Institute Fellowship**, MIT

2017-2018 **BCS Hilibrand Graduate Student Fellowship**, MIT

2016-2017 **Henry E. Singleton(1940) Presidential Fellowship**, MIT

2014 **ECE Chairmans Award**, Volgenau School of engineering, GMU

2012 **Dean Fellowship**, Volgenau School of engineering, GMU

2005,2007,2010 **Honors student in ECE**, IUST